EXERCISE 1: Case study: Connected, but not connected?

At 08:40, three students report that websites do not open on their laptops. Their Wi-Fi icons show "Connected." A fourth student can browse normally using the same access point. All four students can print to a shared network printer. The technician checks one affected laptop: its IP address is 192.168.10.45 and its gateway is 192.168.10.1. A ping to the gateway succeeds. A ping to a known reachable public IP succeeds, but opening a website by its domain name fails. The technician suspects a DNS issue on the affected laptop but does not claim that the router is broken. At 08:55, the technician checks the configured DNS resolver and compares it with a working laptop before making changes.

1.Does "Connected" prove that websites must open?

2. Give TWO observations that support the idea that the local network works.

3. Which evidence suggests a name-resolution issue rather than total loss of IP connectivity?

4. Is it correct to say that the access point is definitely faulty? Justify.

5. Identify one symptom, one observation and one hypothesis from the incident.

EXERCISE 2: Matching: choose the MOST accurate meaning

Match each term (1–10) with ONE definition (A–L). Two definitions are distractors.

1. Switch	A. A possible explanation not yet confirmed
2. Router	B. A device that connects different networks
3. Access point	C. The user-visible sign of a problem
4. Gateway	D. A service that translates names into IP addresses
5. DNS	E. A device that connects local devices and forwards frames
6. Network interface	F. A device that provides Wi-Fi access to a network
7. Ethernet cable	G. A route used to reach other networks
8. Server	H. Hardware or virtual component that connects a device to a network
9. Symptom	I. A wired medium that carries network data
10. Hypothesis	J. A system that provides services or resources

EXERCISE 3: True / False / Not enough evidence

Write T, F or NE (not enough evidence). For every F, correct the statement. The case study is your only evidence.

1. All four laptops have lost access to the local network. ____

Correction (if F): _____

2. A working network printer proves that the internet connection is healthy. ____

Correction (if F): _____

3. At least one laptop can browse websites using the same access point. ____

Correction (if F): _____

4. The gateway is definitely offline because websites fail. ____

Correction (if F): _____

5. The affected laptop can reach its gateway by IP. ____

Correction (if F): _____

6. The router is certainly broken. ____

Correction (if F): _____

7. DNS is a reasonable hypothesis, not a confirmed root cause. ____

Correction (if F): _____

8. The technician changed DNS settings at 08:55. ____

Correction (if F): _____

9. The affected laptop uses an Ethernet cable. ____

Correction (if F): _____

10. A successful public-IP test proves that every website will load. ____

Correction (if F): _____

EXERCISE 4: Complete with the correct technical term

Word bank: gateway • symptom • switch • DNS • interface • troubleshoot • access point • hypothesis • server • connection • router • observation. Use TEN words; two are extra.

1. “Websites do not open” is a reported _____, not a confirmed cause.

2. A _____ provides wireless devices with access to the local network.

3. A _____ connects different networks, whereas a switch connects local devices.

4. The default _____ provides a route beyond the local IP network.

5. A successful printer test is an _____ that the printer is reachable.

6. A _____ is a possible explanation that still requires testing.

7. A network _____ may be physical or virtual.

8. The technician must _____ the issue before replacing equipment.

9. A _____ can provide shared files or other network services.

10. If domain names fail but reachable IP addresses work, check _____ resolution.

Practice: Write two sentences explaining why "no internet" does not necessarily mean "no local network."

_____